

CRF State Summary v8

Pattern Mining + Master System Session — April 2026

Maturity: 8.0/8 (Theory) · 7.5/8 (Empirical gate open)

Ougit Jarittum · ORCID: 0009-0009-4451-6811

April 2026

1 The Primitive Chain

$$\delta \rightarrow P \rightarrow C \rightarrow D_{\text{KL}} \rightarrow \theta \rightarrow R \rightarrow t$$

Everything derives from (ε_0, n, N) : ε_0 from EIC/SINDy, $n = 8$ from SO(3) #16, N enters via extreme-value statistics of the Ψ field. With $n = 2^{d_s}$: $\ln n = d_s \ln 2 = d_s \alpha$.

2 New This Session

ID	Identity	Gap	Significance
K11	$\beta \cdot D_0 = \ln n$	0.005%	Sharpest β identity
K12	$D_0 \cdot H_b = \pi/2$	0.038%	π from K2×K3
K14	$F \cdot \varphi^2 = 4$	0.057%	Golden wave amplitude
K16	$\beta \cdot K^* = \ln(n)/n$	0.005%	Corollary of K11+K1
K17	$\kappa/K^* = \beta/\sqrt{d_s}$	exact	New: K7÷K1
K18	$\beta/\kappa = n\sqrt{d_s}/D_0$	exact	New: K11÷K7
K19	$\beta\kappa = \ln^2(n)/(n\sqrt{d_s}D_0)$	0.005%	New: K11×K7
K7 corrected	$\kappa = \sqrt{d_s} \ln 2/n$	0.047%	K7 formula fixed
p derived	$p = (n+1)/(n+2)$	exact	Dirichlet $n+2$ sources
$n = 2^{d_s}$	Binary Born rule	structural	Key to K11/K17/K18

3 Master System: All Constants from $\ln(n)/D_0$

$$D_0 = n(1 - e^{-1/n}) = 0.94003 \quad [\text{K2}] \quad (1)$$

$$K^* = D_0/n = 1 - e^{-1/n} = 0.11750 \quad [\text{K1}] \quad (2)$$

$$\beta = \ln(n)/D_0 = d_s \alpha / D_0 = 2.2121 \quad [\text{K11}, 0.005\%] \quad (3)$$

$$\kappa = \ln(n)/(n\sqrt{d_s}) = \sqrt{d_s} \alpha / n = 0.15007 \quad [\text{K7}, 0.047\%] \quad (4)$$

All six pairwise relationships exact at 0.005%: $\beta D_0 = \ln n$, $\beta K^* = \ln(n)/n$,
 $\beta \kappa = \ln^2(n)/(n\sqrt{d_s}D_0)$, $\kappa/K^* = \beta/\sqrt{d_s}$, $\beta/\kappa = n\sqrt{d_s}/D_0$, $n\kappa = \sqrt{d_s} \ln 2$

4 Complete Status Table

Result	Status	Gap	Source / Note
<i>Foundations</i>			
δ as primitive	Declared	—	Axiom 0
Born Rule via Gleason	Theorem	—	Session 11
$n = 3$ from $\text{SO}(3)$	Derived	—	#16
$\alpha = \ln 2$	Derived	—	Session 8
Schwarzschild, $G = \ln 2 \cdot c^2/D_0$	Derived	—	Paper v4
<i>Resolution lattice</i>			
$D_0 = n(1 - e^{-1/n})$ (K2)	Derived	0.03%	
$K^* = D_0/n$ (K1)	Derived	0.43%	
$\theta(K^*) = K^*$	Derived	0.001%	
$n = 2^{d_s}$ (binary Born rule)	Physical arg	exact	
<i>Information constants</i>			
$\beta \cdot D_0 = \ln n$ (K11)	Derived	0.005%	New (K11)
$\kappa = \sqrt{d_s} \ln 2/n$ (K7)	Hypothesis	0.047%	Corrected
$\kappa/K^* = \beta/\sqrt{d_s}$ (K17)	Derived	exact	New
$\beta/\kappa = n\sqrt{d_s}/D_0$ (K18)	Derived	exact	New
$\beta\kappa = \ln^2(n)/\dots$ (K19)	Derived	0.005%	New
<i>β chain</i>			
$T_{\text{avg}} = 2nK^*/[(n-1)\varepsilon_0]$	Derived	1.2%	#21a, Wald
$(H_a + H_b)/2 = \sqrt{d_s}$ (K3)	Derived	0.011%	
$\text{Cov}(\psi_{\text{grad}}, \Delta H) = 0$	Derived	exact	K3
β formula (measured inputs)	Derived	0.44%	
<i>Ψ-field</i>			
$p = (n+1)/(n+2)$	Derived	exact	New: $n+2$ sources
$\sigma_\Psi = \varepsilon_0(n+2)/\sqrt{2n+3}$	Derived	0%	AR(1)+ p
$\psi_{\text{grad}} = \sigma_\Psi/\ln \varphi$ (K8)	Hypothesis	0.016%	
$\varphi_{\text{var}} = 1/2 - 1/n^2$	Derived	2.3%	
$F = (1 + \varphi_{\text{var}})(1 + \psi_{\text{grad}})$	Derived	1.1%	

5 Key Numbers v8

$\alpha = \ln 2$	$\varepsilon_0 = 0.00755$	$D_0 = 0.94003$	$\kappa = 0.15007$	$K^* = 0.11750$
$n = 8$	$N = 500$	$d_s = 3.031$	$\beta = 2.212$	$F = 1.527$
$T_{\text{avg}} = 35.4$	$\varphi = 1.61803$	$p = 9/10$	$H_a = 1.7925$	$H_b = 1.6717$
$\varphi_{\text{var}} = 0.4844$	$\sigma_{\Psi} = 0.01732$	$\psi_{\text{grad}} = 0.03600$	$t_1 \approx 300\text{ms}$	$t_2 \approx 485\text{ms}$

6 Open Gaps (Priority Order)

Priority 1: $m(m-1) = 1$ without s_{max}

The energy balance: $m(m-1) = \beta\kappa/f(\delta)$ has $m = \varphi$ as unique solution when $f = \beta\kappa = \ln^2(n)/(n\sqrt{d_s}D_0)$. The function f is now identified as the product of the two known constants K11 and K7. Deriving why the energy balance condition normalises to exactly $f = \beta\kappa$ is the final algebraic step.

Priority 2: $\kappa = \sqrt{d_s} \ln 2/n$ physical proof (K7)

Physical argument is in place: one bit distributed over n modes with $\sqrt{d_s}$ geometric scaling. Formal derivation from Born resampling: show that the IDV-to- Ω transfer per event equals $\sqrt{d_s}\alpha/n$ from first principles.

Priority 3: $n = 2^{d_s}$ formal proof

Physical argument: binary Born rule in d_s spatial dimensions. Formal: derive from the δ -chain that the Born-rule mode count is $n = 2^{d_s}$. This would make K11/K17/K18 all exact from δ .

Priority 4: P0-H Subject 4 (empirical gate)

CRF prediction: flat 50% accuracy across all RT bins (blocked type). This is the empirical gate to Maturity 8.0/8.

7 Documents This Session

Document	Content
CRF_KStar_Closure.pdf	K* gap closed; fcorr derived; D0=1-n exact
CRF_K11K12K14.pdf	K11/K12/K14; $n = 2^{d_s}$ structural key
CRF_MasterSystem_K17K19.pdf	K17/K18/K19; p derived; master chain
CRF_StateSummary_v8.pdf	This document

$\beta = \ln(n)/D_0$. $\kappa = \ln(n)/(n\sqrt{d_s})$. All constants share one numerator: $\ln(n) = d_s \ln 2$.

The system has one number. The number comes from $n = 2^{d_s}$.

2^{d_s} comes from binary resolution in three spatial dimensions.

Three dimensions come from δ .